

Jaswanthi Boyapati

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PROFESSIONAL SUMMARY

MS Data Science graduate and AI Engineer with hands-on experience building and deploying AI systems that solve real business problems. Published 19 production-ready AI applications covering multi-agent orchestration, RAG pipelines, and agentic workflows, all live and publicly accessible. Strong command of the Anthropic Claude ecosystem including Claude Code, MCP, and ADK, alongside vector databases, n8n automation, and LangChain. Comfortable working across teams and helping non-technical colleagues adopt AI tools in their day-to-day work. Looking for a role where I can keep building at the intersection of AI engineering, software development, and business impact.

EDUCATION

University of Houston <i>Master's degree in Data Science (CGPA: 3.96/4.0)</i> • Coursework: Big Data Analytics, Databases, Data Mining, Machine Learning	Houston, TX Jan '23 – Dec '24
Institute of Aeronautical Engineering <i>B.Tech, Computer Science and Engineering (GPA: 9.22/10)</i> • Coursework: Cloud Computing, DSA, Data Warehousing, Object-Oriented Programming, Web Technologies	Hyderabad, India Jun '18 – Aug '22

EXPERIENCE

Libsys <i>AI Engineer</i> • Led the design and deployment of a centralized data and AI platform for a solar monitoring software company, enabling large-scale analysis of inverter performance, site reliability, and portfolio-level operational trends. • Built agentic workflows and automation pipelines using Claude Code, LangChain, ADK, and n8n, helping teams in operations, sales, and analytics use AI in their daily work without needing engineering support. • Developed a sales intelligence system that uses NLP, feature engineering, and classification models to process M&A news and market signals, scoring and prioritizing leads to support the revenue pipeline. • Built end-to-end data pipelines to ingest, clean, and enrich data from operational systems, external market sources, and user activity logs, supporting KPI tracking and downstream machine learning models. • Designed a technician scheduling and routing model using BigQuery and geospatial data, applying optimization and forecasting techniques to reduce travel time and improve allocation efficiency by 85%. • Trained and led a team of interns and junior engineers in data science workflows including data exploration, model development, validation, and deployment while reviewing work and setting technical direction.	Chicago, IL June '25 – Present
University of Houston <i>Research Assistant – Data Science & AI</i> • Built AI agents using the Anthropic Claude API to automate author disambiguation, gender attribution through NamSor, and centrality metric computation, cutting out repetitive data processing and speeding up scholarly network analysis. • Developed an interactive AI chatbot so researchers could query co-authorship metrics, gender disparity insights, and visualizations in plain language, without having to do any manual data exploration. • Tutored 150+ students in SQL, Python, and web development through classroom sessions. Maintained Tableau dashboards with automated Python data ingestion to track grades and attendance across semesters.	Houston, TX Jan '23 – May '25
Client: Chevron Corporation <i>Cloud Data Engineer</i> • Architected and optimized end-to-end ETL pipelines on AWS: landed raw feeds in Amazon S3, ran transformation jobs via AWS Glue, and loaded curated tables into Amazon Redshift, cutting data latency by 30% and improving report accuracy. • Automated pipeline monitoring and alerting with AWS Lambda functions running scheduled data quality checks and configured CloudWatch alarms to flag failures and anomalies, maintaining 99% uptime. • Performed exploratory data analysis and cleaning on raw financial datasets using Pandas and NumPy, embedding validation rules to ensure 100% data integrity for downstream forecasting models.	Bengaluru, India March '22 – Dec '22
Virtusa <i>Data Analyst</i> • Used SQL for data aggregation and manipulation across multiple databases, improving the efficiency and accuracy of organ donor data analysis and producing more reliable insights for decision-making. • Engineered a Python program to automate the collection and processing of organ donor data from Excel spreadsheets across 30+ regions, reducing data collection time by 20% and ensuring more consistent updates. • Helped a client optimize donor outreach using donation analytics visualized in Tableau, supporting more effective campaigns and improving overall donation rates through data-driven strategy.	Hyderabad, India Oct '19 – March '22

PROJECT HIGHLIGHTS

Automated Browser Task Agent <i>Python, Playwright, Claude API, Async Execution, Session Management</i> Built an autonomous browser agent that handles full web workflows on its own, including authentication, dynamic page navigation, form filling from structured data inputs, and logging results to CSV. Designed the whole thing around async execution so multiple tasks can run in parallel, with resilient session management and error handling so it keeps working	
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reliably without supervision. The kind of tool that directly maps to automating ad platform workflows, Shopify data pulls, or any repetitive cross-tool process.

RAG Knowledge Engine

RAG, ChromaDB, Vector Embeddings, Hybrid Search, MMR, Claude API, Streamlit, PyPDF2

Built two RAG pipeline implementations to understand the full spectrum of the architecture. The production version uses ChromaDB as a persistent vector store with three retrieval modes: cosine similarity, hybrid keyword-plus-semantic search, and Maximal Marginal Relevance to reduce redundancy in results. Every answer traces back to its source chunk through citation mode. Supports PDF, DOCX, TXT, and CSV. The second version is a zero-infrastructure in-memory implementation using NumPy cosine similarity, built to show the same RAG concepts without any external dependencies. Both pipelines cover the complete stack from document ingestion and chunking through embedding, retrieval, and grounded generation.

AI Salary Negotiation Coach

Multi-turn Dialogue, Role-Playing Agent, Claude API, Prompt Engineering, Streamlit

Built a salary negotiation tool that covers the whole process rather than just answering questions. It pulls market salary data with percentile breakdowns, scores any offer across five dimensions, and generates negotiation scripts for common scenarios like competing offers, promotion asks, and benefits discussions. The most interesting part is the live negotiation simulator where Claude plays the recruiter with configurable personality types, ranging from firm to flexible to aggressive, so users can practice their counter-offer strategy in a realistic back-and-forth before the real conversation. Good example of using multi-turn dialogue and persona-driven prompt engineering to build something that actually changes how people behave.

Three-Tier VPC Network Deployment Using Terraform

Terraform, AWS VPC, NAT Gateway, Security Groups, Multi-AZ, Infrastructure as Code

Designed and implemented a secure, scalable three-tier Virtual Private Cloud on AWS from scratch using Terraform. The architecture separates presentation, application, and database layers into isolated subnets for security and performance. Automated the provisioning of 10+ components including VPCs, subnets, NAT Gateways, and Security Groups, achieving 99.99% uptime with seamless scaling across two availability zones.

SKILLS

- **Programming:** Python, SQL, JavaScript, Node.js, HTML/CSS, React.js
- **AI and LLMs:** Anthropic Claude API, Claude Code, Prompt Engineering, OpenAI API, Gemini API, OpenRouter, LLM Evaluation
- **Agentic AI:** Multi-Agent Orchestration, Agentic Workflows, LangChain, LangGraph, ADK, MCP, n8n, Tool Use and Function Calling, Autonomous Research Loops
- **RAG and Vector Databases:** Retrieval-Augmented Generation, ChromaDB, FAISS, Pinecone, Hybrid Search, MMR, Document Chunking and Embedding
- **Data Science and ML:** Pandas, NumPy, Scikit-learn, Machine Learning, Neural Networks, NLP, Feature Engineering, ETL Pipelines, Tableau, Plotly
- **Generative AI:** Text-to-Speech (OpenAI TTS), Image Generation (DALL-E 3), Streamlit, Multimodal AI
- **Cloud and DevOps:** AWS (S3, Glue, Redshift, Lambda, CloudWatch), BigQuery, Apache Airflow, Docker, Kubernetes, Terraform, CI/CD, Serverless

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate (SAA-C03) [Link]
- Infosys Certified Software Programmer [Link]

LEADERSHIP & ACTIVITIES

Graduate Indian Student Organization (GISO)

University of Houston

Event Organizer & Committee Member

Jan '23 – Dec '24

- Spearheaded the planning and execution of multiple large-scale cultural festivals and events, coordinating logistics for 3,000+ students including venue selection, vendor negotiations, budget management, and volunteer coordination.
- Managed event promotion through social media campaigns and campus outreach, achieving a 40% increase in student participation compared to previous years and fostering a vibrant cultural community on campus.

Editor-in-Chief, Compodium (Campus Newspaper)

Hyderabad, India

Institute of Aeronautical Engineering

Aug '20 – May '22

- Led an editorial team of 15+ student journalists, photographers, and designers producing a bi-monthly campus newspaper, establishing editorial guidelines and quality standards that elevated the publication's professionalism.

Content Creator & Video Editor

Houston, TX

University of Pittsburgh Partnership Initiative

Sept '24

- Scripted and edited a promotional teaser trailer using Final Cut Pro, combining motion graphics, audio mixing, and narrative storytelling into a 90-second video used across social media platforms and reaching 10,000+ views.

INTERESTS

- **Fitness:** Regular at Orange Theory Fitness where I love the challenge of high-intensity workouts; currently training for HYROX Houston 2027, excited to push my limits in this fitness racing competition.
- **Matcha Making:** Genuinely obsessed with the art of making matcha; love experimenting with different grades, whisking techniques, and recipes from classic usucha to lattes and baked goods.
- **Art:** Love unwinding with canvas painting in my free time; it is my creative outlet where I experiment with colors.